



University of Al-Ameed
College of Medicine
Department of Biochemistry



DIGESTION OF LIPIDS

Assistant prof.
Dr. WEAAM AWAD

3.10.2026

Objectives

- ❖ Study the principal lipids present in the foodstuffs.
- ❖ Digestion of dietary lipids in mouth and stomach
- ❖ Digestion in duodenum and small intestine
- ❖ Study the lipolytic enzymes present in the pancreatic juice.
- ❖ Learn in detail about emulsification of fats, what are Micelles?
How they are formed?
- ❖ Study in details what happens to these hydrolytic products in intestinal epithelial cell after absorption from the gut lumen
- ❖ Study the defects in digestion and absorption of lipids
 - Steatorrhoea
 - Chyluria

- Lipids include a wide variety of chemical substances, such as:
 - Neutral fat (triacylglycerol or triglycerides)
 - Fatty acids and their derivatives
 - Phospholipids
 - Glycolipids
 - Sterols
 - Fat soluble vitamins (A, D, E and K).
- more than 90% fat intake in the diet is as triacylglycerols, with the remainder consisting of cholesterol, cholesterol ester, phospholipid and free fatty acid



Digestion in Stomach

➤ lingual lipase

from the mouth enters stomach along with the food. It has an optimum pH of 2.5-5, It acts on short chain triglycerides which present in milk, butter, The action of lingual lipase is observed to be more significant in the newborn infants.

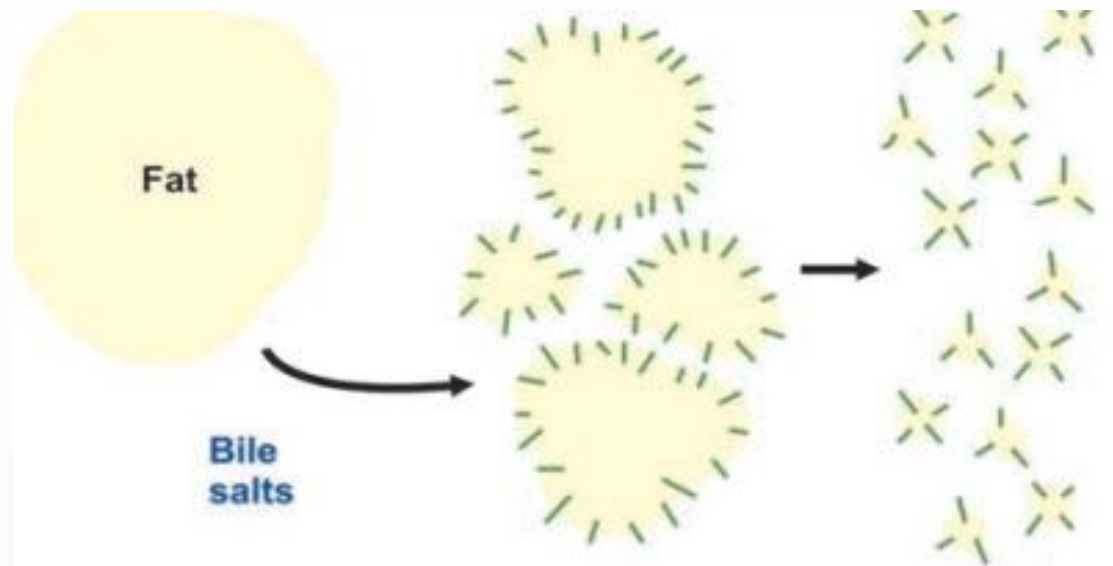
➤ **Gastric lipase**

is acid stable, with an optimum pH around 5.4. Up to 30% digestion of triglycerides occurs in stomach

Emulsification

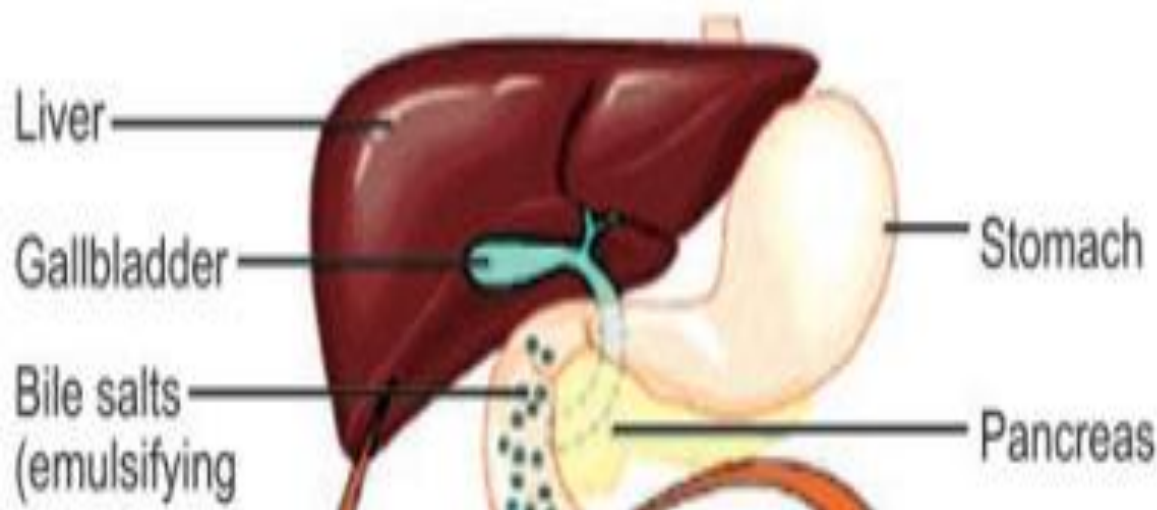
The lipids are dispersed into smaller droplets; surface tension is reduced; and surface area of droplets is increased. This process is favored by:

1. Bile salts (detergent action)
2. Peristalsis (mechanical mixing)
3. Phospholipids



Digestion in Intestines

Bile Salts are Important for Digestion of Lipids ,it lowers the surface tension. They emulsify the fat droplets. The emulsification increases the surface area of the particles for enhanced activity of enzymes



Lipolytic Enzymes in Intestines

1. Pancreatic lipase with Co-lipase
2. Cholesterol esterase
3. Phospholipase A₂.

The bile (pH 7.7) entering the duodenum serves to neutralise the acid chyme from the stomach and provides a pH favorable for the action of pancreatic enzymes.

1. Pancreatic lipase with Co-lipase

Lipase hydrolyses fatty acid in the 1 and 3 positions of the triacylglycerol, producing 2 monoacylglycerols and two molecules of fatty acids, Thus digestion of TAG is partial (incomplete).

2. Cholesterol esterase

hydrolyses cholesterol ester, which produces cholesterol plus free fatty acid

3. Phospholipase A2.

This enzyme catalyzes the hydrolysis of fatty acid residues at the 2-position of the phospholipid, leaving lysophospholipids and a molecule of fatty acid

Intestinal lumen

Intestinal mucosal cell

Apolipoprotein (B-48)

General
circulation

Thoracic duct

Lymphatic
system

2-MAG

2-MAG

Resynthesis

TG

FA

FA

Activation

Thiokinase

Fatty acyl-CoA

C

C

Resynthesis

CE

LysoPL

LysoPL

Resynthesis

PL

Vitamin

Vitamin

Vitamin

A,D,E,K

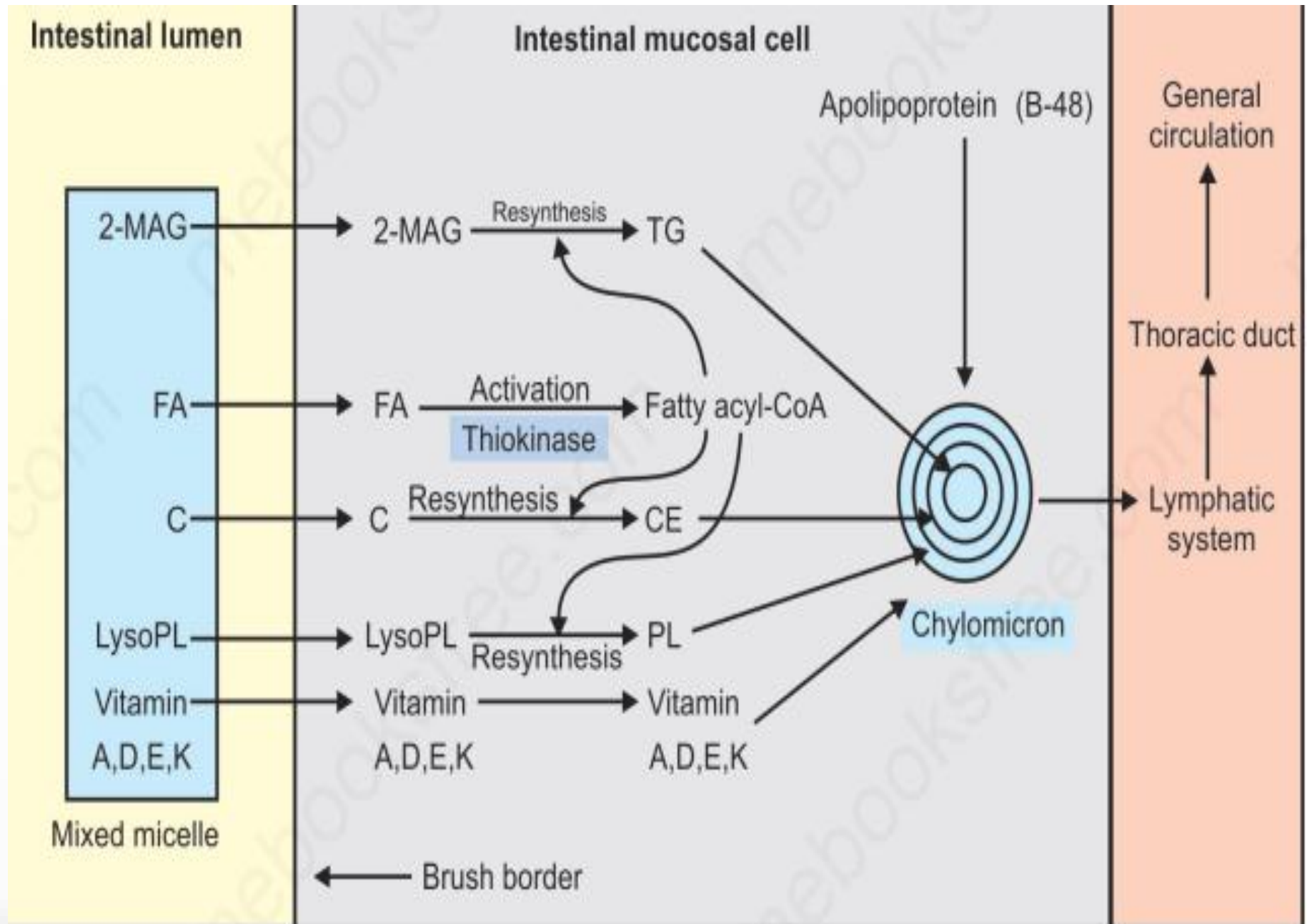
A,D,E,K

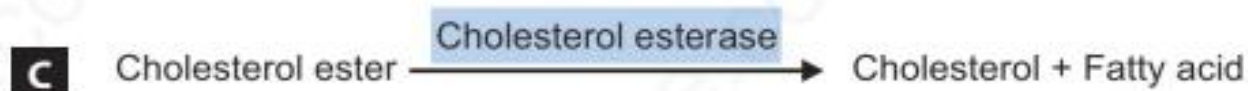
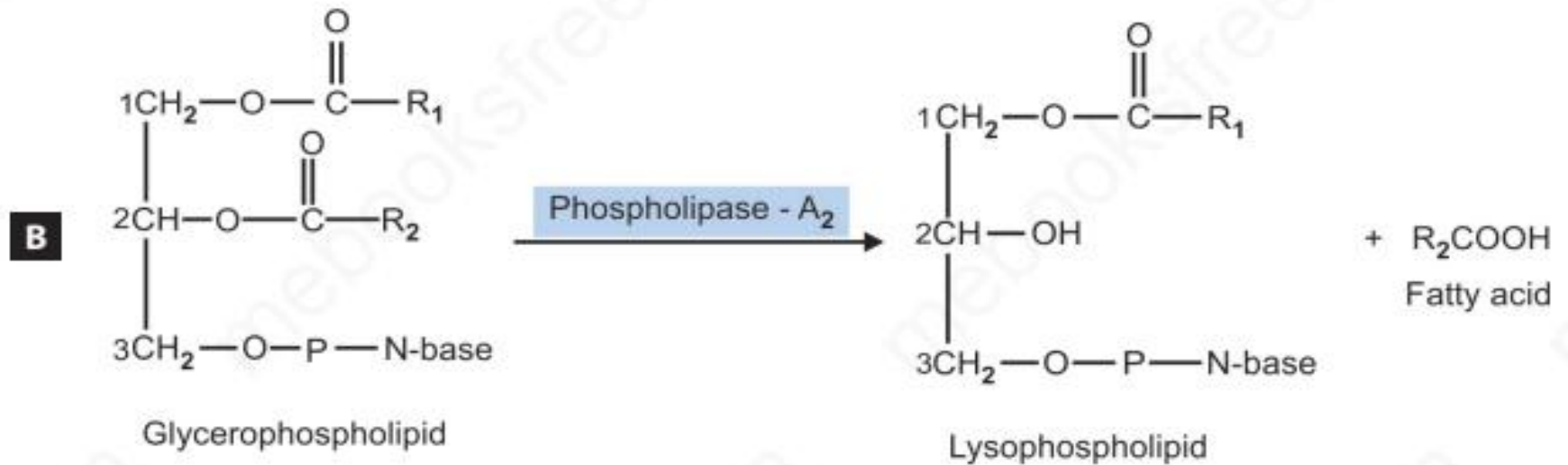
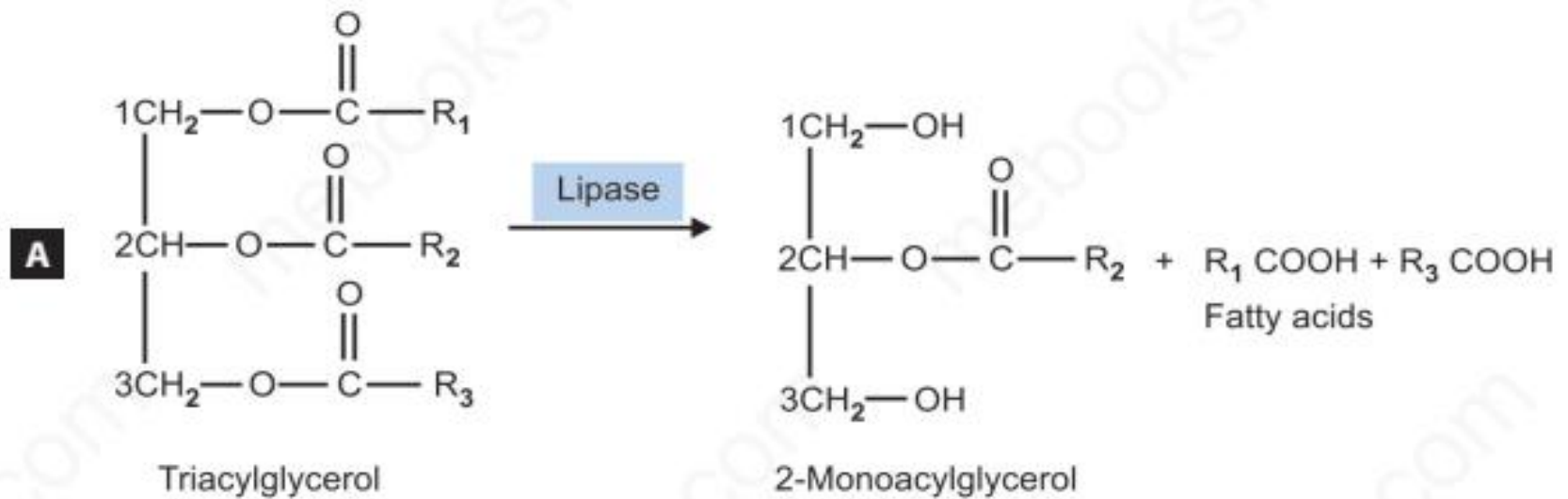
A,D,E,K

Chylomicron

Mixed micelle

← Brush border

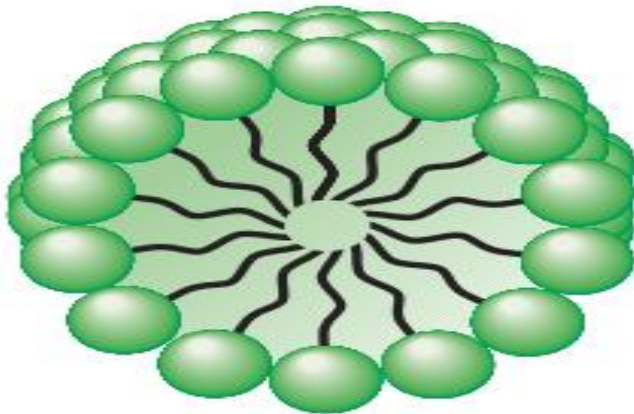




ABSORPTION OF LIPIDS

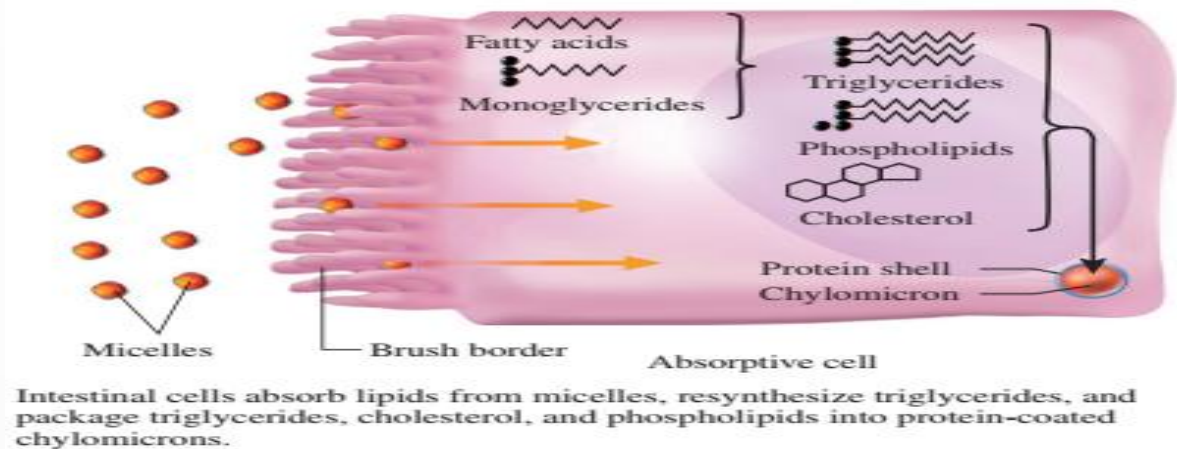
Mixed Micelle Formation

Free fatty acids, free cholesterol, 2-monoacylglycerol, and lysophospholipid are the primary products of dietary lipid digestion. These, together with bile salts, form mixed micelles. Fat soluble vitamins A, D, E and K are also packaged in these micelles and are absorbed from the micelles along with the products of dietary lipid digestion



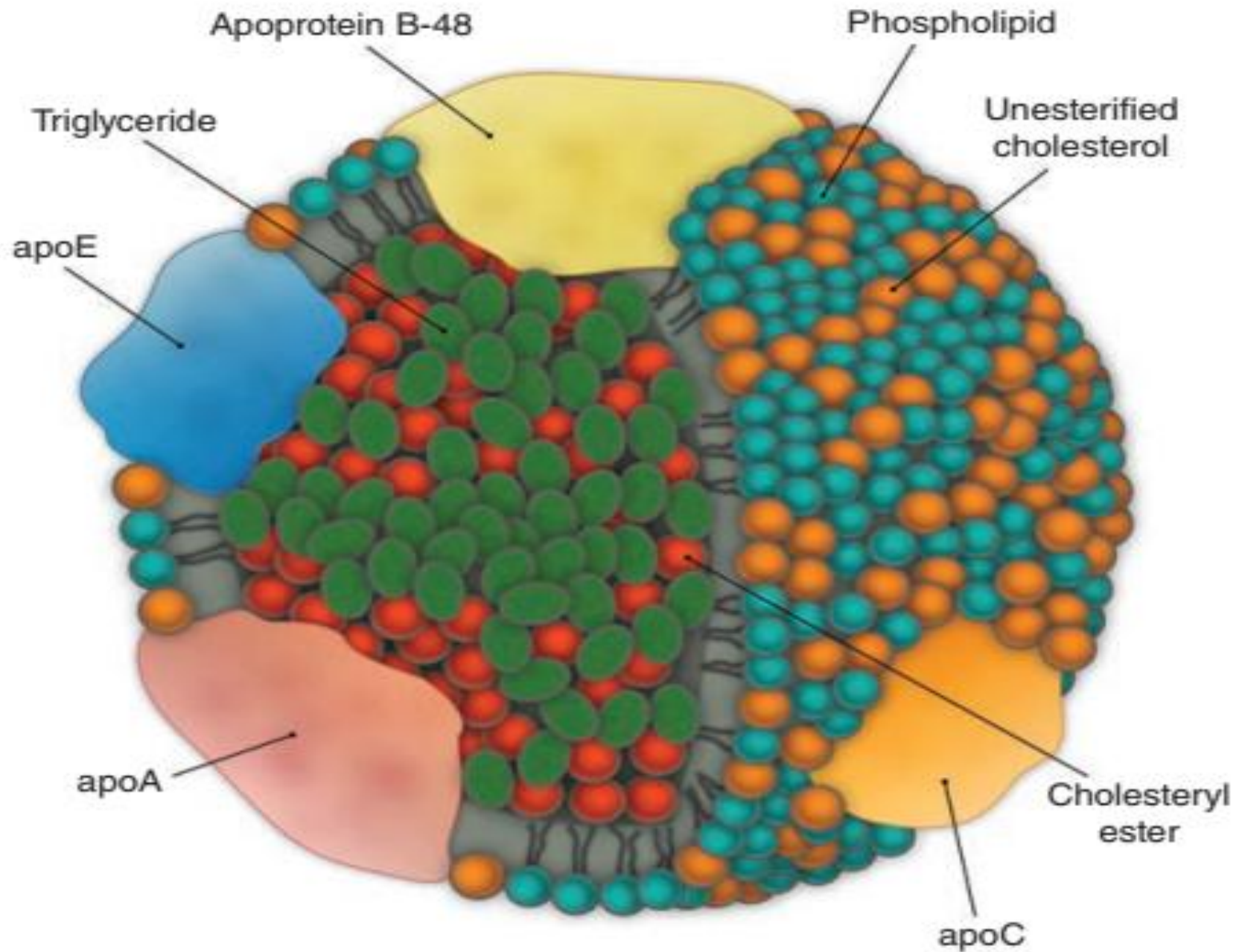
Absorption of Lipids

- ❖ The mixed micelles approach the brush border membrane of the intestinal mucosal cells.
- ❖ There the lipid components from mixed micelles are absorbed into the mucosal cells by diffusion.
- ❖ The net result is the transfer of monoacylglycerol, fatty acids, cholesterol, and lysophospholipid molecules into the mucosal cell



- ❖ Fatty acids are utilized after its conversion to its active form, acyl-CoA
- ❖ 2-monoacylglycerols are reconverted to triacylglycerols.
- ❖ The absorbed lysophospholipids and cholesterol are also reconverted to phospholipids and cholesteroesters.
- ❖ The triacylglycerol resynthesized in intestinal cells combine with cholesterol, phospholipids and proteins to form **chylomicrons**.

Chylomicron

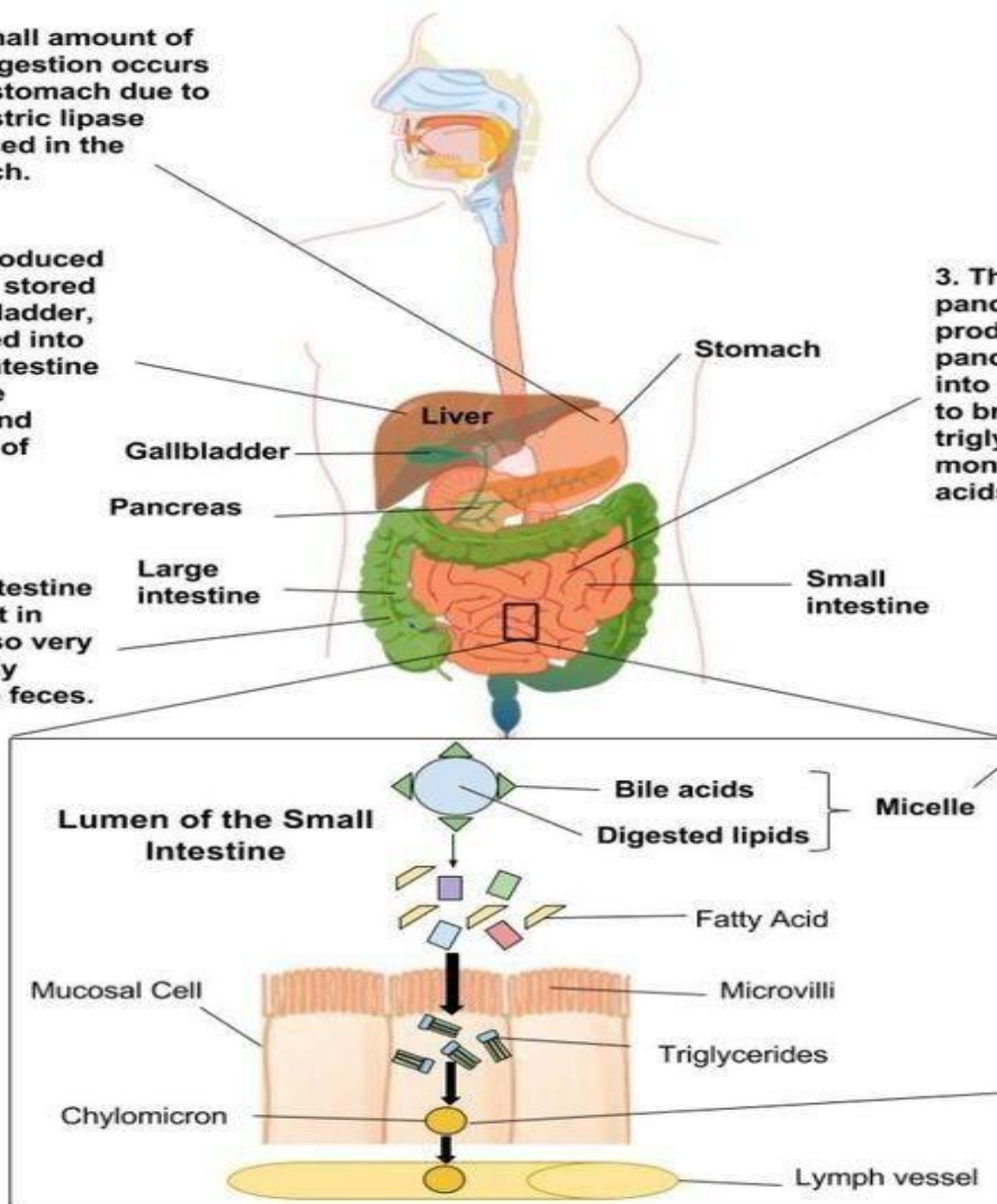


1. A small amount of lipid digestion occurs in the stomach due to the gastric lipase produced in the stomach.

2. Bile is produced in the liver, stored in the gallbladder, and released into the small intestine to aid in the digestion and absorption of lipids.

3. The enzyme pancreatic lipase is produced by the pancreas and released into the small intestine to break down triglycerides into monoglycerides, fatty acids, and glycerol.

6. The small intestine is very efficient in absorbing fat so very little is normally excreted in the feces.



4. In the small intestine, the products from fat digestion and bile acids form a micelle, which moves toward the microvilli to allow the lipids to diffuse into the mucosal cells.

5. Inside the mucosal cells, the fatty acids and monoglycerides are reassembled into triglycerides and incorporated into lipid transport particles called chylomicrons, which enter the lymph vessel.

CLINICAL SIGNIFICANCE

Steatorrhoeas:

can occur due to obstruction to flow of bile due to biliary obstruction, diseases of pancreas, and tropical sprue. This leads to increased fat content of faeces

EXERCISES

1. What effect do salivary enzymes have on triacylglycerols?
2. What are the major products of triacylglycerol digestion?
3. What is the content of a fatty acid micelle?



Q. Determine the location within the human body where each of the following aspects of lipid digestion occurs :

- a. Pancreatic lipases are active.
- b. Fatty acid micelles are produced.
- c. Chylomicrons are produced.
- d. Monoacylglycerols are converted back to triacylglycerols.



